
ML-Approach to Qualimetry: Building Property Trees and Calculating Value Weights using Generative Neural Networks

Abstract

For sustainable Integrated Territorial Development (ITD), evaluating multifaceted value (social, ecological, etc.) is necessary, beyond traditional economic cost. Classical qualimetry for this assessment is labor-intensive and requires numerous experts. This work proposes an ML-based approach, combining classical qualimetric methodology with generative neural networks (GNNs) to automate qualimetry and partially replace experts. A prompt, developed for conducting such hybrid qualimetric studies of territorial value, is presented. It has been tested on test cases (reproducing classical analysis on examples from the works of Prof. Azgaldov G.G.). Experiments confirmed the principal possibility of applying GNNs for automating key qualimetric procedures: building property trees and calculating value weights. Results show that using AI enhances practicality, scalability, and accelerates qualimetric studies in assessing territorial value.

Projects of the integrated development of the built-up territories (CCT) are a powerful comprehension of the transformation of the urban space, aimed at the creature-composed, stable and economically prosperous environment. Traditionally, they are mainly on the economic component - the cost of icapsons. Such an approach focused on measurable financial indicators can lead to a missing from other, no less important aspects that determine the urban environment and the well-being of its inhabitants.

The monograph [6] proposes an approach to the implementation of KRT projects, which is based on the concept of rational and most effective use (RNEI) [5]. This concept is characteristic of the theory and practice of land management, but it is not used very rarely used by developers and urban planners. The concept is based on flooring the value of the territory, and not its value. In contrast to the value of value, the value of the territory should take into account such important aspects as social, environmental, cultural, aesthetic factors, etc. The underestimation of these factors can cite to projects that are effective from an economic point of view, but have negative-headed consequences for the urban environment and its inhabitants, which is critical for sustainability.

The problem of assessing the value of real estate was fundamentally investigated by the Sarah Seis gateway and its colleagues from the school of the cereweigers of the Kingstonnaya University "Assessment of real estate: from value to value" [7].

The combination of various values of stakeholders into a single integral indicator is a difficult task [6], which requires the accounting of various points and the use of methods such as multicriterial analysis and determination coefficients. . Different groups of stakeholders (for example, developers and local livers) can have conflicting ideas about value. The analysis conducted in [6] tried that it is

not possible to make a correct assessment of the value of the territory with instruments of economic measurements [3]. For this procedure, it is necessary to apply the methods of quantitative quality assessment, in particular the method of text.

Qualimetry, scientific discipline engaged in quantitative assessment of the formation [1, 2, 8], was formed back in the 60s of the last century, its active conductor in the practice of measurements was prof. Azgaldov G.G. However, the theory of qualimetry did not receive wide expansion. First of all, due to the complexity of research.

It is laborious and requires the involvement of many experts, often with a separation from Ikhnosnovaya activity for a considerable time. To develop a methodology for evaluating the formation (IOC), the creation of an organizational, technical and expert groups is required. The estimate of the expert group can reach 7-10 people, sometimes creative-haired such groups are required. The above algorithm turned out to be very expensive and laborious, since it was formed without taking into account modern technologies.

The emergence of artificial intelligence (AI), especially generative neural networks (STS), opening a damage to the labor of labor capacity quantitative quality assessment [4]. STS, such as DeepSeek and Gemini 2.0 Flash, demonstrated a high level of training and expertia. STS is able to incite and generate text and code similar to human, as well as execute.

Tasks related to the reasoning and processing of information. They are possessions, relevant qualimetry analysis, including understanding of the text, extraction of information and recognition of images. AI is already used in various analytical tasks and expert areas, such as medical diagnostics or analysis of legal documents. The main problem of the researcher when working with the correct task for the work of the STS, which is called Prompt. The development of effective interactive Promov is a strategy for the STS department in the implementation of qualimetric analysis at the expert level. Properly developed prosts provide context, instructions and restrictions on the model. Despite their capabilities, modern STS have restrictions. They can admit inaccuracies ("hallucinations"), inherit displacements from training data, experience difficulties with complex numerical reasoning and do not have true beings or common sense. The translation of multifaceted judgments and implicit knowledge of experts into explicit rules for AI is a significant problem and for a time cannot be fully automated and require the verification of the transmission data of the model.

To work out the qualimetric Prompt among the Monica STS integrator with the sincoling of the DEPSEEK V3 STS, a bot named "Kalimetry" was created. Work on qualimetry was loaded into the knowledge base, first of all, the works of G.G. Azgaldova. The first Prompt was developed to fulfill the procedures for compiling trees of the properties and conservation of calculations of weights of simple properties. Simple properties are the properties that can be measured analytically or expertly, and their weights are defined as a tanning coefficient (Y.N.K.).

To debugging Prompt and confirming the possibility of using the STS, an experiment was conducted with the reproduction of an example from the book of prof. Azgaldova G.G. [2, 8] on the creation of a qualimetric methodology for determining the laureates of the National Franchising Award of the Golden Brand in the Golden Franchise nomination (as of the state of 2007). In the original example and complexity of work, it was estimated at about 9 people-days, excluding the costs of the head of the development of the methodology. Based on 19, the criteria was built, a tree of properties, represented in Table 1. The numbering of the properties of a warginal example was not applicable for the work of the bot, which is required to change the encoding.

Table ??.

Next, a bot training was carried out in interactive mode, where errors were bothewed, and the relevant instructions were added to the PrOMT to prevent them. Thus, it was possible to train the bot to carry out the correct calculations of the lerteology of qualimetry. The final Prom amounted to 16,380 signs (including symbols., Smaching).

After receiving satisfactory training results, the bot was given to the downtime to reproduce an example from the book of prof. Azgaldova G.G. With interactive use with the bot, they revealed a certain discrepancy between the vine of the properties of the rules of composing such trees developed. Nevertheless, he was completely reproducing an example from the book and calculate the weight coefficients of the rules of qualimetry, which he did. The results of a comparison of the output parameter (Y.N.K.) and the values calculated by the bot is presented in Table 2

Table 1: Property tree from the example of efficiency analysis of franchise-nominated enterprises from the book by Azgaldov G.G.

Level 0	Level 1	Level 2	Level 3	Level 4	Level 5
Franchise efficiency	18. Results obtained - 1.1.	16. Organizational work on franchises - 1.1.1.	13. Generalized franchise characteristics - 1.1.1.1.	12. Prevalence by number of franchises - 1.1.1.1.1.	1. Number of franchisees with multiple franchises - 1.1.1.1.1.1. 2. Total number of franchisees - 1.1.1.1.1.2.
				3. Regional prevalence - 1.1.1.1.2.	
			14. Franchise agreement success rate - 1.1.1.2.	4. Number of terminated franchises - 1.1.1.2.1.	
				5. Percentage of terminated franchises - 1.1.1.2.2.	
		6. Total sales volume - 1.1.2.			
	19. Generalized cost characteristics - 1.2.	17. Franchising costs - 1.2.1.	7. Franchisor's expenses on franchise advertising - 1.2.1.1.		
			15. Franchisor's expenses on franchise promotion - 1.2.1.2.	8. Costs for franchisee support - 1.2.1.2.1.	
				9. Sunk costs from franchise termination - 1.2.1.2.2.	
			10. Goodwill enhancement costs in franchise advertising - 1.2.1.3		
		11. Share of franchising costs in total advertising expenses - 1.2.2.			

90 As can be seen from the table. 2, the results obtained by the bot are very close, for example, screaming.
91 Some inconsistency is explained by the fact that in the example of the value of the group coefficients,
92 they were averaged according to experts, while the bot appointed nimbly. It is noteworthy that the
93 average data of experts in some cases were used to prostrate the rules of qualimetry. For example, in
94 the decomposition of the parental property, "The success of the conclusion of the franchises" (Table
95 3), the experts gave directly opposite assessments, which may indicate questions for their competence.
96 In such cases, the rules of the second round of assessment are required to coordinate the positions of
97 experts, which was not made, which led to an error in the calculations of the example. In this delay,
98 the calculation of the bot was more correct and relevant to the Rules.

99 Thus, in the first approximation, it was proved that the bot can correctly, a screw -mode, create
100 trees of properties and conduct the necessary calculations of qualimetry corrective. This conclusion
101 made it possible to proceed to assess the value of the territories. For this goal, a bot in another
102 advanced version of the Gemini 2.0 Flash corporation Google was created. The created Prompt was
103 transferred to the “qualimetryGM” bot and the materials on the theory and practice of qualimetry
104 and the assessment of the territories. The bot was the task to build a tree of properties and make
105 calculations of weight coefficients (Y.N.K.), provided that the three basic properties of value are
106 equal to: economic, environmental and social. The interactive regime was turned off, and the bot
107 should have been sagged to the parental properties to simple, measured, and manufacturing. To solve
108 the problem of the bot, it was sufficient to decompose for 3 yarus (Table 4).

109 Conclusion

110 1. Although the STS has significant potential for transformationQualimetric analysis, a complete
111 replacement of expert people at this stage is presented with malae.To reproduce the nuances of human
112 judgment, implicit knowledge, common sense to and create complex, contextual-dependent situations.
113 The most likely script. The future is a hybrid approach in which the STS is used as a powerful
114 tool for complementing and expanding the capabilities of expert people. AIS canotomatize rulfinas,
115 processed -large -lubricantsProvide initial results or offers (for example, quality criteria for weight).
116 This allows experts to focus on more complex, strategic shoes of analysis, requiring a deep objective
117 understanding and critical thinking.

118 2. Development of effective interactive Promov plays a key roleIn the direction of AI models to
119 perform the tasks of qualimetric analysis with high flow and relevance. ITERIVE PROMES and
120 Feedback cycles, possibly the sector of a person-expert, can help in improving work.

121 3. The integration of the STS into qualimetry raises important ethical andPractical questions. It is
122 necessary to clarify the issues of accountability for errors, ensure the transparency of assessments
123 based on AI, and take into account the risk of perpetuating or displacement of displacements from
124 training data. Practical aspects include accessibility and the quality of training data, the necessary
125 computing resources and training -users. The growing role of AI will require skills adaptation experts,
126 including production development, testing models and solving complex ethical dilemmas. It is
127 possible to maintain supervision and testing from a person in hybrid approaches to enforce accuracy,
128 reliability and ethical use.

129 4. Accounting for multifaceted value along with the economic cost isThe key factor in the success and
130 long -term sustainability of projects of complex development of territories. The scymetry.quantitative
131 assessment of various aspects of value and their integration in the process of receiving decisions.
132 The article shows and proves the fundamental possibility of II in the form of the STS for conducting
133 research and measurements using the qualimetry. The following studies should be aimed at develop-
134 ingpractical methods for assessing and integrating value into the practice of qualimetric measurements
135 of value.

Table 2: Comparison of Qualimetric Analysis Results: Example vs. Bot Calculation

Indicator	Tree Node	Example from Book			Code	Bot W
		Expert Avg.	Norm. Gi	Calculated YNK		
Franchisees with multiple franchises	1	74	0.426	0.0637	1.1.1.1.1.1	
Total franchisees	2	100	0.574	0.0858	1.1.1.1.1.2	
Regional distribution	3	77	0.435	0.1153	1.1.1.1.2	
Terminated franchises	4	51	0.456	0.0720	1.1.1.2.1	
Termination ratio	5	61	0.544	0.0860	1.1.1.2.2	
Total franchise sales	6	47	0.324	0.2022	1.1.2.	
Franchisee support costs	8	100	0.778	0.0972	1.2.1.2.1	
Contract termination losses	9	29	0.222	0.0278	1.2.1.2.2	
Franchisor marketing expenses	7	60	0.290	0.0937	1.2.1.1	
Goodwill enhancement share	10	67	0.324	0.1048	1.2.1.3	
Franchising cost share in marketing	11	16	0.138	0.0516	1.2.2.	

Note: YNK = Tiered Normalization Coefficient; Verification for Tier 5 = 1.0000

Table 3: Franchise Success Rate

Indicator	Tree Node	Non-Normalized Group Importance Coefficients (Experts)							Mean
		Exp.1	Exp.2	Exp.3	Exp.4	Exp.5	Exp.6	Exp.7	
Terminated franchises	4	10	100	10	20	20	100	100	51
Termination ratio	5	100	10	100	100	100	10	10	61

Table 4: Economic Value Indicators for Comprehensive Territory Development

Tier 0	Tier 1	Tier 2	Tier 3	WNC
Territory Value for CDP	Economic Value (100%)	Local Economy Diversity (100%)	Retail facilities availability	0.0289
			Service sector development	0.0260
			Food service representation	0.0231
			SME presence	0.0202
		Business Infrastructure (90%)	Office spaces/coworking	0.0327
			Telecom quality	0.0294
			Business service accessibility	0.0261
		Transport Connectivity (80%)	Public transport network	0.0231
			Pedestrian infrastructure	0.0208
			Cycling infrastructure	0.0184
			Road accessibility	0.0161
		Job Creation Potential (70%)	Available business spaces	0.0256
			Entrepreneurial appeal	0.0230
			Labor market alignment	0.0205

Designed in the article “Claramimetry” “You are an expert on qualimetry and build a tree of properties for solving a definite problem. It is necessary to build a tree of properties according to the laws of qualimetry of the peremns until the simple properties that can be measured. 1. Always ask the user at the beginning of the dialogue: "Remind you of the rules for compiling the properties according to the methodology of prof. Korostelev SP". Wait for the user's reaction! You will continue only after the user's reaction/no → if “no” go to paragraph 2, and if “yes”, we display the text: “Rules for creating properties of properties” 10 key rules for compiling trees of properties according to the methodology: Determination of the main parental property (GRS) 1. GRS is an initial problem or a goal that is on tier 0. It always develops a weight of 100

8. The use of expert estimates of the purpose of weights and verification of the correctness of the tree involves experts with knowledge in the subject area. 9. Documentation of changes in the change in the tree of properties is recorded with the date and cause of the correlation. This ensures transparency and traceability of the process. 10. A check for completeness and consistency of the core should cover all aspects of the problem being studied and not maintain protesters between properties on different tiers: 10.1. The coverage of all key aspects of properties should include all the significant aspects of the problem under study, such as functionality, reliability, ease of use, cost and others. For example, for a CRM system, this can include not only its functionality, but the interface, speed of work, the cost of maintenance and other important characteristics. 10.2. The exclusion of contradictions by all levels of the tree must exclude the contradictions between the properties. For example, if one property implies high cost, and the other is low, lays coordination and clarifications. Contradictions can also arise between the properties of the worldly levels, therefore it is important to check their relationship and logical consistency. 10.3. The check for completeness should be complete, i.e.

Table 5: Environmental Value Indicators

Tier 0	Tier 1	Tier 2	Tier 3	WNC
Territory Value for CDP				
	Environmental Value (100%)	Environmental Quality (100%)	Air pollution level	0.0289
			Water quality	0.0260
			Soil contamination	0.0231
			Noise pollution	0.0202
		Green Spaces (90%)	Green area per capita	0.0327
			Green zone coverage	0.0294
			Green space quality	0.0261
		Biodiversity (80%)	Flora diversity	0.0260
			Fauna diversity	0.0289
			Protected species	0.0231
		Energy Sustainability (70%)	Renewable energy share	0.0256
			Building energy efficiency	0.0230
			Resource use efficiency	0.0205

Table 6: Social Value Indicators

Tier 0	Tier 1	Tier 2	Tier 3	WNC
Territory Value for CDP				
	Social Value (100%)	Social Infrastructure (100%)	Education facilities	0.0265
			Healthcare facilities	0.0265
			Culture/sports facilities	0.0239
			Social welfare facilities	0.0212
		Safety (90%)	Crime rate	0.0327
			Perceived safety	0.0294
			Law enforcement efficiency	0.0261
		Social Cohesion (80%)	Public spaces	0.0291
			Community activity	0.0262
			Citizen participation	0.0233
		Living Comfort (70%)	Maintenance quality	0.0144
			Housing affordability	0.0160
			Housing quality	0.0144
			Utility infrastructure	0.0128

158 Cover all the necessary properties for the solution of the task. This includes both the properties of
159 the purpose of the object and its functional, operational and other characteristics. For example, for a
160 tent, this can include protection against moisture, the strength of the material, the convenience of
161 installation and other aspects. 10.4. Accounting for all aspects of the problem of the properties of
162 the properties should take into account all the key aspects of the problem under study. This is both
163 the properties of the purpose of the object and its functional, operational and other characteristics.
164 For example, if an object is a CRM system, a tree is not only to use its functionality, but also the
165 convenience of use, reliability, service cost and other important aspects. 10.5. The exclusion of the
166 contradictions of all tiers of the tree must exclude the contradictions between the properties. For
167 example, if one property implies high cost, and the other is low, we currently coordinates approval
168 and clarification. Contradictions can also arise between the properties of a worldly tiers, therefore
169 it is important to check their relationship and logical consistency. These rules provide a systematic
170 approach to building trees and Ichispolization to solve multi -criteria problems. ”

171 Always turn on all points and subparagraphs of the rules, including 10.1, 10.2 and so on. 2. Now
172 we need to decompose the complex integral property that is located to the tier 0 and does not have
173 the encoding defined in the name of the problem. To clarify the context of the task before the
174 decomposition: to clarify the context of the task and priority aspects before the decomposition
175 of the integral property in order to recover the discrepancies in the substantiation of the weights.
176 School 1: request the user’s context of the task. Specify which aspect is a priority for the task:
177 economic, environmental, social or other? This will help to correctly determine the key factors for
178 decomposition. "Step 2: Offer decomposition options based on an updated context. Example:" If
179 the economic aspect is a priority, the key factor will be economic attractiveness. Fix the selected
180 approach. Example: “Selected approach: Economic attractiveness as a key factor. Fixing date:
181 2025-04-02.” Checking data: Make sure that the context of the task and priority aspects are agreed by
182 the Spain. Care that the selected approach is recorded and used for all subsequent steps. The format
183 of the conclusion: an updated context of the problem. Approach to decomposition. Fixing of the
184 approach. Example: enter the context of the task: “Evaluation of the territory for investment.” Specify
185 the priority aspect: “Economic.” Select the approach: “Economic attractiveness as a key factor.”
186 Complement the approach: “Selected approach is recorded. Date: 2025-04-02.” Error messages:
187 error 1: Error 1: Error 1: Error 1: Error 1: Error 1 "The context of the problem is not specified. Please
188 specify the priority aspect." Error 2: "The selected approach is not fixed. Please fix the approach by
189 continuation."

190 Recommendations: The selected context always take into account before the start of the decompo-
191 sition. Fix the selected approach and use it for all subsequent steps. If the context of the problem
192 changes, create a new decomposition based on updated dummies. Excerpt “how many quasi properties
193 you need to decomp integration” and present the user for approval. And after the answer, the
194 performance and assignment of the property with the property (for the first tier, for example, 1.1 and
195 1.2 for the separation of 2 quasi properties) and, after consistent with the user, weight - by the name
196 of the value of 100

197 For each simple property, check the chain of normalized weights from it to the GRS. Reduce the
198 standardized weights in the chain and get Y.N.K. Carry and correct errors with the standardized
199 weightation: ensure the correct use of normalized weights when constructing the properties and
200 calculation of Y.N.Ski execution: checking the standardized weights: if the normalized weight is
201 equal 1.0000, this indicates an error, since this is possible for the GRS (tier 0). For all other tiers,
202 the standardized weight should be less than 1.0000, and the summary of all quasi -free properties of
203 the parental property should be equal to 1.0000. Marking: if a normalized weight equal to 1.0000,
204 recount the formula: make sure that the amount of normalized weights All quasi -free properties of
205 parental care is 1.0000. I.N.K. For all simple properties, using a chain of normalized scales from a
206 simple property to the GRS. Crossing the amount of Y.N.K.: Make sure that the amount is Y.N.K.
207 All simple properties are 1.0000 ± 0.01 . Example use: Make sure that the final table with scales
208 and I.N.K. Fixed cannot be changed. Before building a tree of properties, check that all normalized
209 weights and Ya.N.K. are used to the rules. Documentation: if you need to make changes, create a new
210 table and a tree of properties. Summarize all Y.N.K. And check if the amount is $1.0,000 \pm 0.01$. If the
211 amount does not match, look for errors in the calculations or data. If the accuracy is unattainable,
212 then bring all the calculations to the convenient table for verification in Excel - in all the calculated
213 values, except for the code, put the comma instead of the point. 2. When working with a tree of
214 properties, strictly observe the following rules: do not change the title of properties: all properties of

the properties should remain, as they are agreed and provided by the user. START DATA: Before making changes or adding new properties, check their suzeres with the consistent data. Cook changes: if the user makes changes, save them in a separate section or table with a indication of the date and table indicating the date and table. The reasons for the adjustment.

Checking for conflicts: if the discrepancy is detected between the current il -secured data, display a message indicating the conflict and requesting a confirmation of amendments. Documentation: All changes and updates record in the table with an indication: tier property of property of the property (

3. If accuracy is achieved, then bring the final results in the form of a readable table with citizens, columns and lines. In the first column, indicate the tier, in the second, the number in the third is the name of the property, in the third weight as a percentage, in the fourth -minute weight value. After each confirmation by the Customer: Save the table in the format: Property Code Violes of Venile (

Formatting: Borders of tables: | And-for dividers. Exposure: texts-on the left edge, number-by the right. Before making changes: Check new parameters with saved data. Example of conflict:* Current parameter: 1.1.1.2.1-"Number of franchise" (weight: 80

PrOMT to preserve the final table of table: the final table is stored in the memory as an unchanged object. All these tables (tiers, codes of properties, names, weights, normalized weights, Ya.N.K. Istatus) are fixed in the current state. Barrow of changes: Any attempts to change the table will deviate changes, the system must derive, the system must withdraw Message: "The final table is recorded and cannot be changed. If you requires the changes, create a new table."

Access to the table: The user can request the output of the table at any time. The table is displayed in the format specified by the user (for example, with dividers | and

- for borders). Construction of integrity: before fixing the table, the system checks that the amount Y.N.K. It is 1.0000 ± 0.01 . If the accuracy is not observed, the system displays the message: "The amount of Y.N.K. is not equal to 1.0000. Please check the data before fixing the table."

In addition to the final table, it is necessary to create another table only with simple. Columns: "Number in order, code, name, Y.N.K., Ya.N.K. Privates." . Before the output of the final table and the truncated table after each kodanado, put the point. Divide the numbers in Y.N.K. In all tables and Y.N.K. In percentage, replace instead of a point on a comma. You need to display these two tables to Excel. Approximately ask the user; "Do I need to convert the final table into the structure of themindmap? Yes/no." If the answer is yes, then: "1. ** Purpose: ** Transform the final table of properties into the MindMap structure with a complete decomposition of branches and verification of data correctness.

2. ** Fulfillment steps: *- ** Step 1: ** import the final table in CSV or Excel format.- ** Step 2: ** Build the hierarchical structure of the mindmap, starting from the root node (tier 0).- ** Step 3: ** for each tier:

Add properties in accordance with their codes and weights.- Make sure that all branches are laid out to simple properties.- ** Step 4: ** Check the correctness of the data:- The sum of normalized weights on each tier should be $1.0,000 \pm 0.01$.- the amount of Y.N.K. All simple properties should be $1.0,000 \pm 0.01$.- ** Step 5: ** Visualize the MindMap structure using tools (for example, minister, XMind, Coggle).3. ** Data verification: *- for each simple property, check the chain of normalized weights from it of the end of root node.- change normalized weights in the chain and get Y.N.K.- summarize everything Y.N.K. And make sure that the amount is $1.0,000 \pm 0.01$.4. ** The output format: *- Mindmap structure in text format with retreats and symbols “ “ ‘.- table with data verification (tier, property code, property name, weight (

Provide the correct use of scales from the final table in the construction of the MindMap structure.Steps of execution:Step 1: Before starting work, make sure that the final table with Libra and Y.N.K. K. K.The is refined and cannot be changed.

Step 2: When building mindmap, strictly use weights and Ya.N.K. From the final table. Irce them yourself. Shag 3: Check that all properties and their weights correspond to the data from the table. Shag 4: Make sure that the amount is Y.N.K. All simple properties are $1.0,000 \pm 0.01$. Shag 5: if a discrepancy is detected, stop the process and inform the error. Monitoring points:

The control point 1: The weight of all properties must strictly correspond to the final table. The control point 2: the amount of Y.N.K. All simple properties should be 1.0000 ± 0.01 . The control

point 3: If the final table is changed, create a new mindmapon structure based on updated data.
Example use:

Enter the final table in the CSV or Excel format. Lower the process of building mindmap, strictly following the data from the table. Care that all weights and Ya.N.K. correspond to the table. The messages about the error:

Error 1: "The inconsistency of the scales was found. Please check the data in the final tightness."
Error 2: "The amount of Y.N.K. is not 1.0000. Please check the data with the transfixation of the table." Recommendations: Fixing data: Make sure that the final table is recorded and cannot be changed: Before constructing mindmap, check that all the weights are all weights and that all weight Y.N.K. Corresponding to Tabled. Documentation: If you need to make changes, create a new table and the MindMap renewal

References

- [1] Azgaldov G. G. *Theory and Practice of Product Quality Assessment: Fundamentals of Qualimetry*. Moscow: Ekonomika, 1982. 256 p. (In Russian)
- [2] Azgaldov G. G., Kostin A. V., Sadovov V. V. *Qualimetry for Everyone*. Moscow: InformZnanie, 2012. 165 p. (In Russian)
- [3] Korostelev S. P. *Theory and Practice of Economic Measurements in Real Estate*. <https://www.litres.ru/book/sergey-pavlovich-kor/teoriyai-praktika-ekonomicheskikh-izmereniy-v-nedvizh-68845437/> (Accessed: 2023-11-21). (In Russian)
- Application of Artificial Intelligence for Modeling Expert Opinion*
- Pp. 36–46. (In Russian)
- [5] Korostelev S. P., Tsypkin Yu. A., Korostelev Yu. S. *Conflict Between Most Efficient and Rational Use in Territorial Development Projects*. International Scientific-Technical Journal "Real Estate: Economics, Management", 2024, Vol. 4 (Supplement), Pp. 190–197. (In Russian)
- [6] Korostelev S. P., Tsypkin Yu. A., Sharipov S. A. *Theory and Practice of Land Management in Urbanized Territories*. Moscow: MIRAA Publishing, 2025. 294 p. (In Russian)
- [7] Sayce S., Smith D., Cooper R., Venmore-Rowland P. *Real Estate Appraisal: From Value to Worth*. Moscow: Russian Society of Appraisers, 2009. 504 p. (In Russian, translation)
- [8] Azgaldov G. G., Kostin A. V., Padilla Omiste A. *The ABC of Qualimetry Toolkit for Measuring the Immeasurable*. Ridero, 2015. 167 p. (In English)

301 Reproduction: The work describes the proposed ML approach that combines Classic qualimetric
302 methodology using the STS. It is mentioned that the conduct of research of the value of the territory
303 was developed by the PrMPT, which is the OSTOBANNENTROURAL

304 Confirming the possibility of using the STS for automation of key procedures: building trees of
305 properties and calculating values. It is described by experiment with a reproduction of an example
306 from the book of prof. Azgaldova G.G. The compares of the qualimetric analysis from the example
307 and calculations made by the bot (table2) is shown. It is indicated that for the development of
308 Prombte, the environment of the integrator of the SGNSA DEPSEEK V3 SGNS integrator was used,
309 and to assess the value of the territories - the Gemini 2.0 Flash12 STS. The bosom of the bot was
310 loaded with work on qualimetry, first of all, by the works of G.G. Azgaldov, and materials on theory
311 and practical -qualimetry and valuation of territories were added to evaluate the territories. The final
312 Prompt amounted to 16,380 signs and presented the application.

313 Transparency: sources of data for training or the knowledge base of bots are disclosed - ecuphors
314 on qualimetry, works of G.G. Azgaldova and materials on theory and practice and assessment of
315 territories. The conclusions of the article explicitly indicate the need to deliberate the transparency of
316 assessments based on AI, as an important ethical Ipractic issue. In the work presented, there are no
317 potential conflicts of interests, financing of work from any sources except their own did not seem to
318 be.

319 Ethics: The article emphasizes the importance of assessing multifaceted value (social, environmental,
320 etc.) in addition to the traditional economic value for sustainable complex development of territories.
321 The underestimation of these factors can lead to a casual, but negative projects from a long -term
322 point of view. This is the awareness of the social and environmental impact of KRT projects. The
323 integration of the STS in qualimetry raises important ethical and practical issues. The book includes:

324 The need to clarify the issues of accountability for errors.

325 The risk of perpetuating or enhancing displacements from training data. This is directly binded
326 with the risk of discrimination, although the work does not detail how this can manifest itself in the
327 assessment of the territories.

328 The need to take into account complex ethical dilemmas in the growing role of A

329 In conclusion, it is noted that the accounting of multifaceted value is a key factor of long -term
330 stability of KRT projects, which corresponds to the ethical approach to Kgorod development.

331 Reliability: the reliability of the approach using the STS is checked by producing classic examples of
332 qualimetric analysis. The results obtained by the bot are compared with the results from the example
333 (table 2) for demonstrating intimacy. It is noted that in one case, the calculation of the bot was more
334 correct and more consequent of the rule of fireimalimeters, which is average for the average person,
335 containing an error due to violation of the rules. This may indicate the potential of JISSENTENTY
336 OF Reliability by strict adherence to the methodology. However, the author is also guided by the
337 restrictions of modern STS, such as the possibility of inaccuracies ("hallucinations") and difficulties
338 with complex numerical reasoning. It is emphasized that the results of the STS require the verification
339 of the output data of the model. It is important to maintain supervision and testing by a person in
340 hybrid approaches to ensure accuracy, reliability and ethical use.

341 Human participation: Classical qualimetry requires the involvement of many experts, which is time
342 -consuming. The ML approach with the STS is proposed for the automation of qualimetry and partial
343 replacement of experts. However, the author directly argues that the complete replacement of expert
344 people of the Nada stage seems unlikely. The most likely scenario is the future of a hybrid approach
345 in which the STS is used to supplement and expand the abilities of expert people. AI automates
346 routine tasks and provides the first results, allowing experts to focus on more complex adversary
347 spaces. The development of effective interactive industrials and feedback cycles, possibly with the
348 participation of a person-expert, can help improve the work of the STS.

349 The growing role of AI will require skills adaptation from expert people, including verification and
350 the solution of ethical dilemmas. The importance of human supervision and verification of Viihbrid
351 approaches to ensure accuracy, reliability and ethics is emphasized by the control. Unlike studies
352 where experiments are conducted with people

353 Participants to study their behavior or reaction, this work describes the application of AI for the
354 performance of tasks that traditionally fill out experts.
355 Directly with experiments on people participating people, visitors are not considered.